

Computer Science & Information Technology

Computer Networks

DPP: 1

IPv4 Address

- Q1** Consider a class-less IPv4 network where network id bits assigned to this network are “01010011 10011011 0110”, identify correct decimal representation of network address which is used to represent this network ?
(A) 83 . 155 . 0 . 0
(B) 83 . 155 . 96 . 0
(C) 83 . 155 . 96 . 255
(D) 83 . 155 . 111 . 255
- Q2** Consider a class-less IPv4 network where network id bits assigned to this network are “01010011 10011011 0110”, identify correct decimal representation of broadcast address which is used to broadcast a packet in this network ?
(A) 83 . 155 . 0 . 0
(B) 83 . 155 . 96 . 0
(C) 83 . 155 . 96 . 255
(D) 83 . 155 . 111 . 255
- Q3** Consider a class-less IPv4 network, the network address of this network is “205 . 215 . 104 . 0” and size of network identifier field is 22 bits, which one of the following is a broadcast address of this network ?
(A) 205 . 215 . 104 . 0
(B) 205 . 215 . 104 . 255
(C) 205 . 215 . 107 . 255
(D) 205 . 215 . 111 . 255
- Q4** Consider a class-less IPv4 network, the broadcast address of this network is “25 . 15 . 95 . 255” and size of network identifier field is 22 bits, which one of the following is a network address of this network ?
(A) 25 . 15 . 64 . 0
(B) 25 . 15 . 80 . 0
(C) 25 . 15 . 92 . 0
(D) 25 . 15 . 95 . 0
- Q5** Consider a class-less IPv4 network, the network address of this network is “140 . 190 . 208 . 0”, which is/are can be a broadcast address of this network ?
(A) 140 . 190 . 208 . 63
(B) 140 . 190 . 223 . 0
(C) 140 . 190 . 223 . 63
(D) 140 . 190 . 223 . 255
- Q6** Consider a class-less IPv4 network, the broadcast address of this network is “120 . 120 . 159 . 255”, which is/are can be a network address of this network ?
(A) 120 . 120 . 136 . 0
(B) 120 . 120 . 144 . 0
(C) 120 . 120 . 159 . 0
(D) 120 . 120 . 159 . 192
- Q7** Consider an IPv4 address of a class-less IPv4 network is “15 . 20 . 25 . 30 / 21”, identify correct pair of first and last IP address of this network which can be assigned to hosts ?
(A) 15 . 20 . 24 . 0 and 15 . 20 . 31 . 255
(B) 15 . 20 . 24 . 0 and 15 . 20 . 31 . 254
(C) 15 . 20 . 24 . 1 and 15 . 20 . 31 . 255
(D) 15 . 20 . 24 . 1 and 15 . 20 . 31 . 254



- Q8** Consider a class-less IPv4 network, the number of bits in the network id field are 26, how many maximum number of hosts can be in this network ?
- Q9** Consider a class-less IPv4 network, the network mask of this network is “255 . 255 . 254 . 0”, how many maximum number of hosts can be in this network ?
- Q10** Consider network mask of a class-less IPv4 network is “255 . 255 . 240 . 0”, identify which pairs of IP addresses could belong to this network ?
(A) 210 . 130 . 148 . 12 and 210 . 130 . 168 . 30
(B) 210 . 130 . 148 . 14 and 210 . 130 . 158 . 32
(C) 210 . 130 . 158 . 16 and 210 . 130 . 168 . 34
(D) 210 . 130 . 164 . 18 and 210 . 130 . 168 . 38
- Q11** Consider two computers A and B and their IP addresses are “110 . 120 . 159 . 125” and “110 . 120 . 149 . 191” respectively. Which network mask they can use if they belongs to same network ?
(A) 255 . 255 . 224 . 0
(B) 255 . 255 . 240 . 0
(C) 255 . 255 . 248 . 0
(D) 255 . 255 . 252 . 0
- Q12** Consider a class B IPv4 network with 6 bit subnetting, how many maximum number of hosts can be in each sub-network ?
- Q13** Consider a class C IPv4 network with 4 bit subnetting, identify correct subnet (sub-network) mask?
(A) 255 . 255 . 255 . 224
(B) 255 . 255 . 255 . 240
(C) 255 . 255 . 255 . 248
(D) 255 . 255 . 255 . 252
- Q14**

Consider a class B IPv4 network and wishes to form subnets for 128 departments, identify correct subnet (sub-network) mask ?

- (A) 255 . 255 . 128 . 0
(B) 255 . 255 . 192 . 0
(C) 255 . 255 . 254 . 0
(D) 255 . 255 . 255 . 0

- Q15** Consider a class C network with four bit subnetting, the network address of the network is “203 . 204 . 205 . 0”. Which is/are can be a valid sub-network (subnet) address of a sub-network belongs to this network ?

- (A) 203 . 204 . 205 . 8
(B) 203 . 204 . 205 . 16
(C) 203 . 204 . 205 . 24
(D) 203 . 204 . 205 . 32

- Q16** Consider a class B network with three bit subnetting, the network address of the network is “150 . 140 . 0 . 0”. Which is/are can be a valid pair of subnet address and subnet broadcast address of a sub-network (subnet) ?

- (A) 150 . 140 . 64 . 0 and 150 . 140 . 127 . 255
(B) 150 . 140 . 96 . 0 and 150 . 140 . 127 . 255
(C) 150 . 140 . 128 . 0 and 150 . 140 . 191 . 255
(D) 150 . 140 . 160 . 0 and 150 . 140 . 191 . 255

- Q17** Consider a subnetted class C network, the broadcast address of a subnet is “220 . 210 . 210 . 31”, which is/are can be a subnet mask of this subnet ?

- (A) 255 . 255 . 255 . 128
(B) 255 . 255 . 255 . 192
(C) 255 . 255 . 255 . 224
(D) 255 . 255 . 255 . 240

- Q18** Consider a class-less IPv4 network with four bit subnetting, how many maximum number of hosts will be reduced in the network after subnetting ?
[Note : Subnet all zero and all one are allowed]



- Q19** Consider an Internet Service Provider (ISP) with an available IP addresses space is “25 . 28 . 144 . 0 / 21”. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B ?
- (A) 25 . 28 . 144 . 0 / 22 and 25 . 28 . 148 . 0 / 23
 (B) 25 . 28 . 144 . 0 / 22 and 25 . 28 . 150 . 0 / 23
 (C) 25 . 28 . 148 . 0 / 22 and 25 . 28 . 144 . 0 / 23
 (D) 25 . 28 . 148 . 0 / 22 and 25 . 28 . 146 . 0 / 23

- Q20** Which one of the following CIDR prefixes exactly represents the range of IP addresses “170 . 120 . 200 . 0” to “170 . 120 . 207 . 255” ?
- (A) 170 . 120 . 200 . 0 / 20
 (B) 170 . 120 . 200 . 0 / 21
 (C) 170 . 120 . 200 . 0 / 22
 (D) 170 . 120 . 200 . 0 / 23

- Q21** Consider routing table of a IPv4 router shown below :

Destination	Mask	Interface
80.82.0.0	255.255.0.0	1
80.82.64.0	255.255.192.0	2
80.82.80.0	255.255.240.0	3
80.82.84.0	255.255.252.0	4
Default		5

Consider an IPv4 datagram contains destination IP address “80 . 82 . 84 . 86” arrive at the router. On which interface will it be forwarded ?

- Q22** Consider routing table of a Classless Inter-domain Routing (CIDR) IPv4

router shown below :

Destination	Interface
220.215.0.0/16	1
220.215.192.0/18	2
220.215.200.0/20	3
220.215.210.0/16	4
Default	5

Consider an IPv4 datagram contains destination IP address “220 . 215 . 210 . 205” arrive at the router. On which interface will it be forwarded ?

- Q23** Address Resolution Protocol (ARP) query always _____ in the network.
- (A) Unicast (B) Multicast
 (C) Broadcast (D) Anycast

- Q24** Supernetting is a process of _____.
- (A) Dividing a network into smaller sub-networks
 (B) Combine multiple networks into a larger one
 (C) Increasing the number of hosts per subnet
 (D) Reducing the number of hosts per subnet

- Q25** NAT (Network Address Translation) is used for _____.
- (A) Dynamic IP address allocation
 (B) Creation of subnets within a network
 (C) Multiple devices to share a single public IP address
 (D) Find MAC address corresponds to given IP address



Answer Key

Q1 (B)
Q2 (D)
Q3 (C)
Q4 (C)
Q5 (A, D)
Q6 (B, C, D)
Q7 (D)
Q8 62~62
Q9 510~510
Q10 (B, D)
Q11 (A, B)
Q12 1022~1022
Q13 (B)

Q14 (C)
Q15 (B, D)
Q16 (B, D)
Q17 (C, D)
Q18 30~30
Q19 (A, B, C, D)
Q20 (B)
Q21 4~4
Q22 2~2
Q23 (C)
Q24 (B)
Q25 (C)



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Hints & Solutions

Q1 Text Solution:

Network Address: 83.155.96.0

01010011 10011011 01100000 00000000

20 bit Net ID 12 bit Host ID

Q2 Text Solution:

Broadcast Address: 83.155.111.255

01010011 10011011 01101111 11111111

20 bit Net ID 12 bit Host ID

Q3 Text Solution:

Network Address: 205.215.104.0/22

205.215.01101000.00000000

Net ID 22 bit Host ID 10 bit

Broadcast Address: 205.215.107.255

205.215.01101011.11111111

22 bit Net ID 10 bit Host ID

Q4 Text Solution:

Broadcast Address: 25.15.95.255/22

205.15.01011111.11111111

Net ID 22 bit Host ID 10 bit

Network Address: 25.15.92.0

205.15.01011100.00000000

22 bit Net ID 10 bit Host ID

Q5 Text Solution:

Network Address: 140.190.208.0

140.190.1101 0000.00000000

Min. 20 bit Net ID Max. 12 bit Host ID

A. 140.190.208.63

140.190.11010000.00 111111

26 bit Net ID 6 bit Host ID

B. 140.190.223.0

140.190.110.111111.00000000

C. 140.190.223.63

140.190.11011111.00 111111

6 bit Host ID

D. 140.190.223.255

140.190.11011111.111111

20 bit Net ID 12 bit Host ID

Q6 Text Solution:

Broadcast Address: 120.120.159.255

120.120.100 11111.11111111

Min. 29 bit Net ID Max. 13 bit Host ID

A. 120.120.136.0

120.120.10001000.0000 0000

11 bit Host ID

B. 120.120.144.0

120.120.10010000.0000 0000

20 bit Net ID 12 bit Host ID

C. 120.120.159.0

120.120.10011111.0000 0000

24 bit Net ID 8 bit Host ID

D. 120.120.159.192

120.120.10011111.11 000 000

26 bit Net ID 6 bit Host ID

Q7 Text Solution:

IP Address: 15.20.25.30/21

15.20.00011001.00011110/21

21 bit Net ID 11 bit Host ID

First Host IP: 15.20.24.1/21

15.20.00011000.00000001/21

Last Host IP: 15.20.31.254/21

15.20.00011 111.1111110/21

Q8 Text Solution:

Net ID: 26bit

Host ID: (32 – 26) = 6 bit

Maximum number of hosts can be in this network

= (2⁶ – 2) hosts

= 62

hosts

Q9 Text Solution:

Net Mask: 255.255.254.0



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255.255.11111110. 0000 0000
 23 bit Net ID Host ID 9 bit

Maximum number of can be in this network = $(2^9 - 2)$ hosts
 = 510 hosts

Q10 Text Solution:

240 → 1111 0000
 148 → 1001 0100
 168 → 1010 1000
 158 → 1001 1110
 164 → 1010 0100

Q11 Text Solution:

IP_A: 110.120.1001 1111.125
 IP_B: 110.120.1001 0101.191
 Maximum 20 bit in netmask

Q12 Text Solution:

Class B:
 Net ID → 16 bits
 Subnet ID → 6 bits
 Host ID → $(32 - 16 - 6) = 10$ bit
 Maximum number of hosts can be in each subnet
 = $(2^{10} - 2)$ hosts

1022 hosts

Q13 Text Solution:

Subnet Mask:
 255.255.255. 1111 0000
 Net ID 4 bit 4 bit
 subnet ID Host ID
 255.255.255.240

Q14 Text Solution:

Class B:
 Net ID → 16 bits
 Number of subnet = 128 department
 Subnet ID → $\lceil \log_2 (128) \rceil$ bits = 7 bits
 Subnets Mask: 255.255.254.0

255.255.11111110. 00000000
 Net ID 7 bit 9 bit
 subnet ID Host ID

Q15 Text Solution:

Subnet Address: 0000
 Subnet Host

8 → 00001000

16 → 00011000

24 → 00011000

32 → 00100000

Q16 Text Solution:

A. 64 → 010 00 00 00
 127 → 011 11 11 11
 B. 96 → 011 00 00 00
 127 → 011 11 11 11
 C. 128 → 100 00 00 00
 127 → 101 11 11 11
 D. 160 → 101 00 00 00
 191 → 101 11 11 11

Subnet Address: 150.140. ---- 00000. 0000
 0000

Broadcast Address: 150.140. ---- 11111. 1111 1111

Q17 Text Solution:

Class: C
 Subnet Broadcast Address: 220.210.210.31
 220.210.210. 000 1111
 Net ID Min. 36bitMax. 5bit
 subnetting Host

Subnet Mask can have 27 to 30 one's

Q18 Text Solution:

4 bit subnetting
 Reduction in number of hosts after subnetting =
 $[(2^4) \times 2 - 2] = 30$ hosts

Q19 Text Solution:

Available IP Address: 25.28.144.0/21

25.28.10010...../21

↓

org.A. 10010000.00000000/22

25.28.144.0/22

org.B. 10010100.00000000/23

25.28.144.0/23

ISP 10010110.00000000/23

25.28.150.0/23

25.28.10010---.-----/21

org.A. 10010100.c

25.28.148.0/22

org.B 10010000.00000000/23

25.28.144.0/23

ISP 10010010.00000000/23

25.28.146.0/23

Q20 Text Solution:

$\underbrace{170.120.11001000.0000\ 0000}_{21\ \text{bit}}$
 $\underbrace{170.120.11001111.1111\ 1111}_{21\ \text{bit}\quad 11\ \text{bits}}$
 $\left. \vphantom{\begin{matrix} 170.120.11001000.0000\ 0000 \\ 170.120.11001111.1111\ 1111 \end{matrix}} \right\} 2^{11}$

CIDR Pre fix.

170.120.200.0/21

Q21 Text Solution:

84 → 010 10100

252 → 111 11100

Q22 Text Solution:

210 → 110 100 10

252 → 111 111 00

208 → 110 100 00

210 → 110 100 10

240 → 111 100 00

208 → 110 100 00

Q23 Text Solution:

Address Resolution Protocol (ARP) query
always Broadcast in the network.

Q24 Text Solution:

Supernetting is a process of combine multiple
networks into a larger one.

Q25 Text Solution:

NAT (Network Address Translation) is used
for multiple devices to share a single public IP
address.



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